External Portal

When a client is redirected to an external portal, the AP will inject a doubly-encrypted, signed, and time-stamped token as a GET parameter to the redirect URL. This token should be returned to the Alta API to authorize a client's temporary access to the network. However, the external portal can also decode this token itself to determine the MAC address (and timing validity, SSID, etc.) of the client.

All fields are encrypted using the IAPP key, except for the Alta Site ID and Encrypted Payload fields, which are encrypted using the Alta SHA256 key. In reality, the Encrypted Payload key is encrypted twice, first with the IAPP key which results in its own value, and then the overall SHA256 key for both the Alta Site ID and Encrypted Payload, inclusive.

This token contains several data types in TLV format, as defined below:

Enumeration Number	**Type**	**Encrypted with**
1	Alta Site ID	Alta SHA256
2	Encrypted Payload. **Length is big-endian 2 bytes instead of 1.**	Alta SHA256
3	Alta SSID ID	IAPP Key
4	Alta Network (Virtual AP Interface starting at index 1)	IAPP Key
5	MAC address (binary)	IAPP Key
6	Timestamp (Big-endian Unix Epoch 32-bits)	IAPP Key

TLV Format:

1 byte - Type	1 byte - Length	(Length) bytes - value

There are two keys used to encrypt and sign the token. The first is the Alta SHA256 key:

```
static const uint8_t alta_sha256[] =
{ 0x9b, 0x12, 0xf4, 0xd4, 0xaf, 0xf3, 0x1d, 0x90, 0xa2, 0x31, 0xaf, 0x2e, 0x90, 0x61,
  0xe9, 0xcd, 0x6c, 0x51, 0x6e, 0xbd, 0x37, 0x19, 0x45, 0xe0, 0xc5, 0x53, 0x3a, 0x18,
  0xc5, 0x91, 0xb3, 0x26 };
```

The second is the site-specific "IAPP Key". This can be obtained from the AP's /cfg/config.json file, under the system.iappkey field. This key is generated when a site is created, and does not change. It should be converted from hexadecimal into binary before being used.

As an example, with an SSIDID of “abcd”, the SSIDID_TLV would look like (in hex octets):
0x03 0x04 0x61 0x62 0x63 0x64

The + operator indicates concatenation. The AES_SIV function is the AES GCM SIV function as defined here: https://en.wikipedia.org/wiki/AES-GCM-SIV

Token:
AES_SIV(SiteID_TLV+PAYLOAD_TLV, Alta SHA256 Key)

PAYLOAD_TLV:
AES_SIV(SSIDID_TLV+NETWORK_TLV+MAC_TLV+TIMESTAMP_TLV, IAPP Key)

Of course since TLV fields are used, the order can/may be rearranged depending on the implementation.

An example Javascript implementation of decoding the token can be found in the wifiAuthEx.js script.